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Nagashima

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(54) **IMAGE FORMING APPARATUS**

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G03G 15/08 (2006.01)

(52) **U.S. Cl.**
CPC ***G03G 15/80*** (2013.01); ***G03G 15/0872*** (2013.01)

(58) **Field of Classification Search**
CPC . G03G 15/0872; G03G 15/0886; G03G 15/80
USPC 399/90, 107, 108, 110
See application file for complete search history.

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(57) **ABSTRACT**

The image forming apparatus includes an image carrier, a developing unit, and a housing. The developing unit includes a toner carrier, a development container, a circuit board, and a board retaining portion. The development container rotatably supports both end portions of a rotating shaft of the toner carrier, and contains developer and the toner carrier. The development container includes a protrusive support portion which protrudes from both end portions of an axial direction and which is pivotably supported by a retaining frame. The development container is pivoted on the protrusive support portion serving as a fulcrum, and the toner carrier is reciprocatively movable so as to come into contact with or separate farther from the image carrier. The board retaining portion is pivotably supported on a fulcrum given by a retention support portion coaxial with the protrusive support portion, and is independent of the reciprocative movement of the development container.

6 Claims, 12 Drawing Sheets

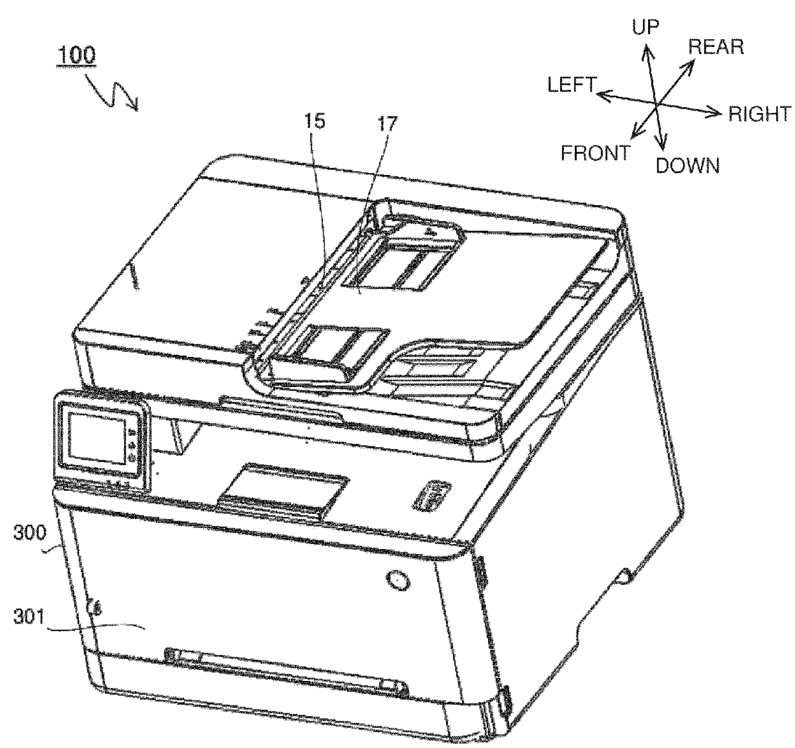


Fig.1

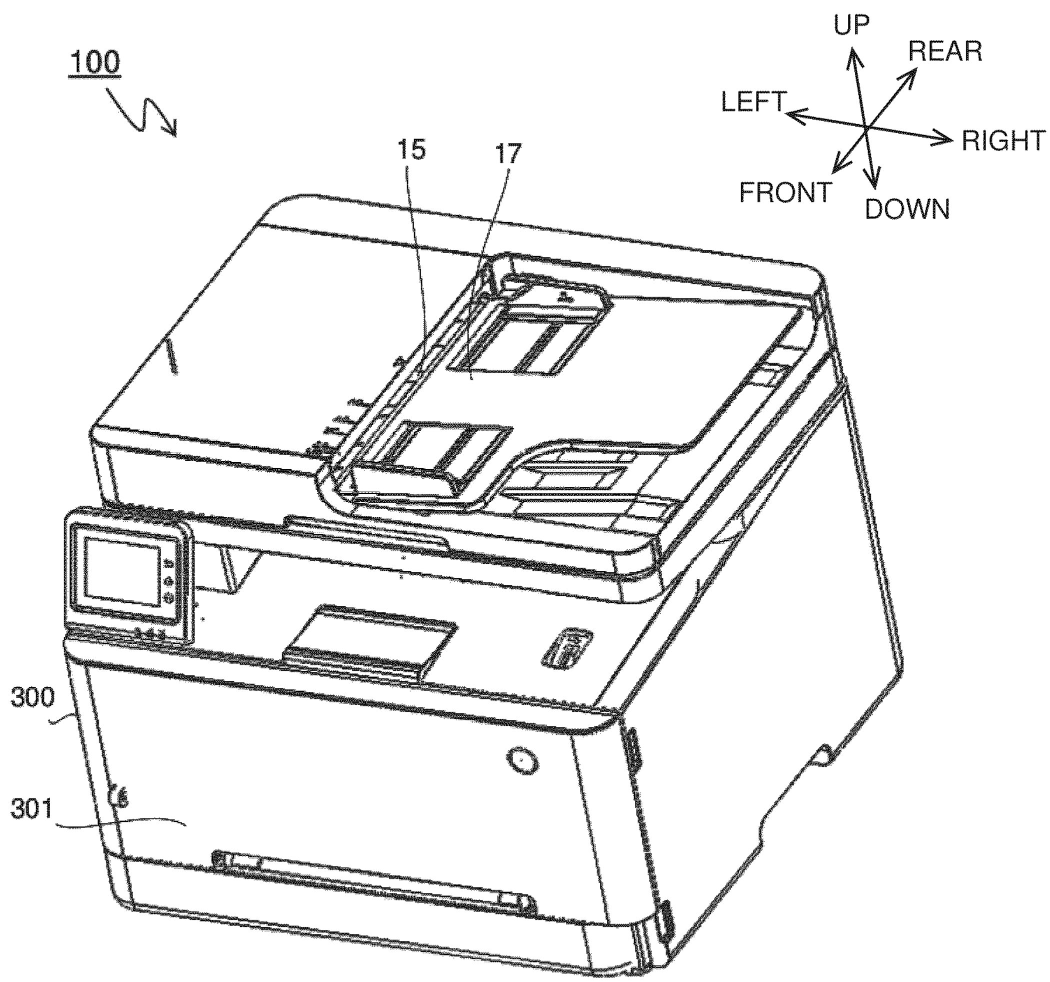
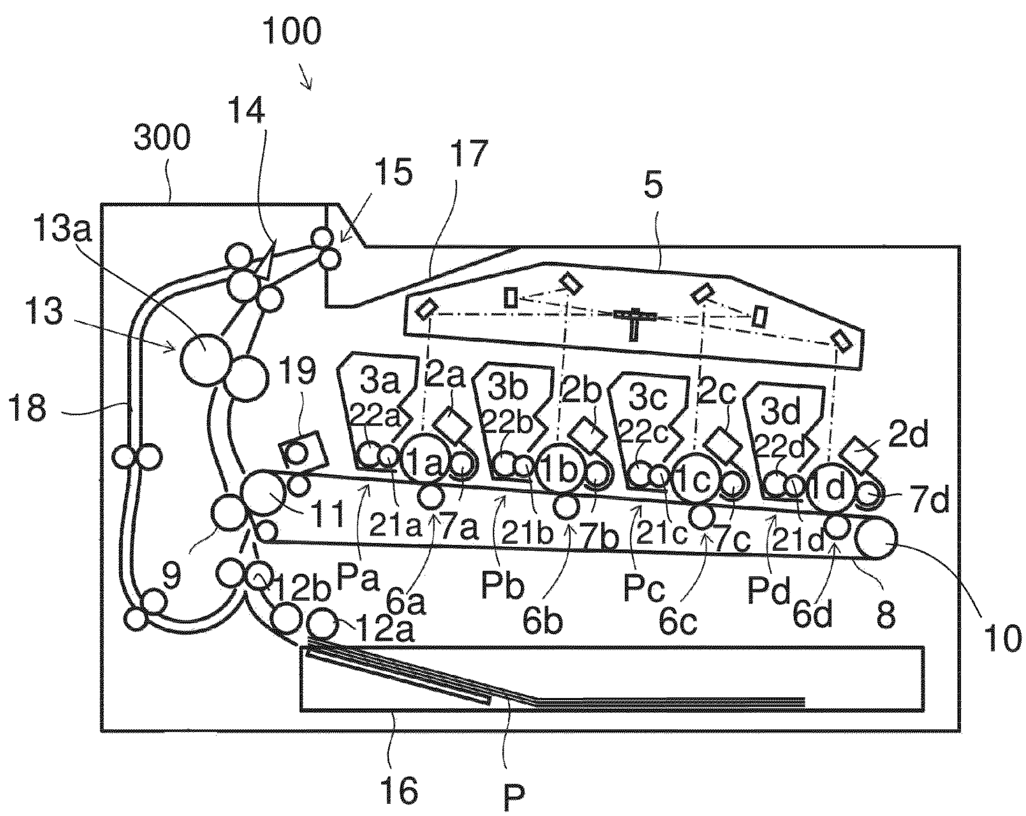


Fig.2



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300
14
15
17
5
13a
13
18
19
3a
2a
22a
1a
3b
2b
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3c
2c
22c
1c
3d
2d
22d
1d
7d
11
21a
7a
21b
7b
21c
7c
21d
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16
P

Fig.3

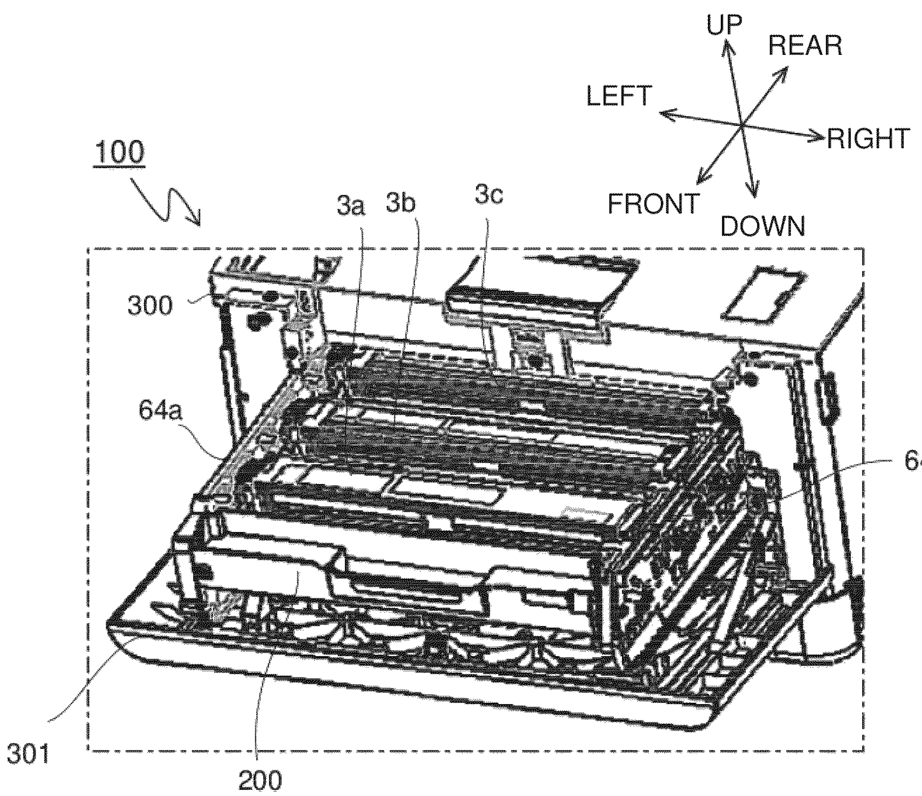


Fig.4

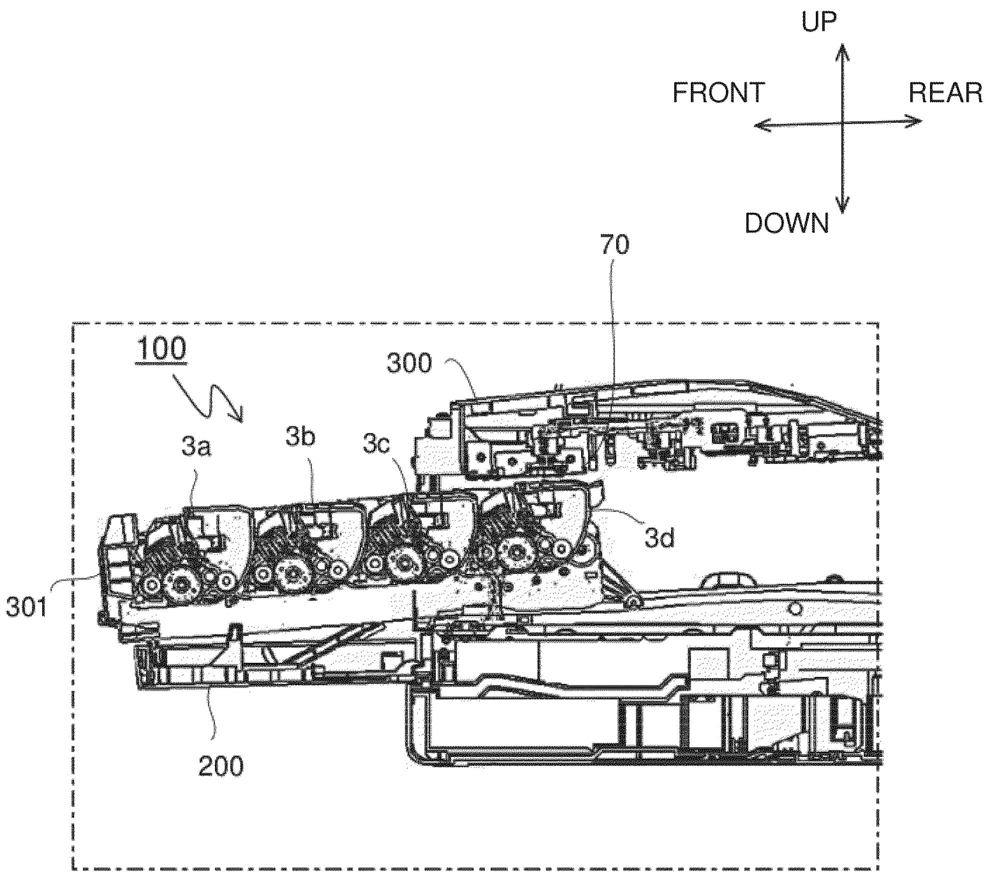


Fig.5

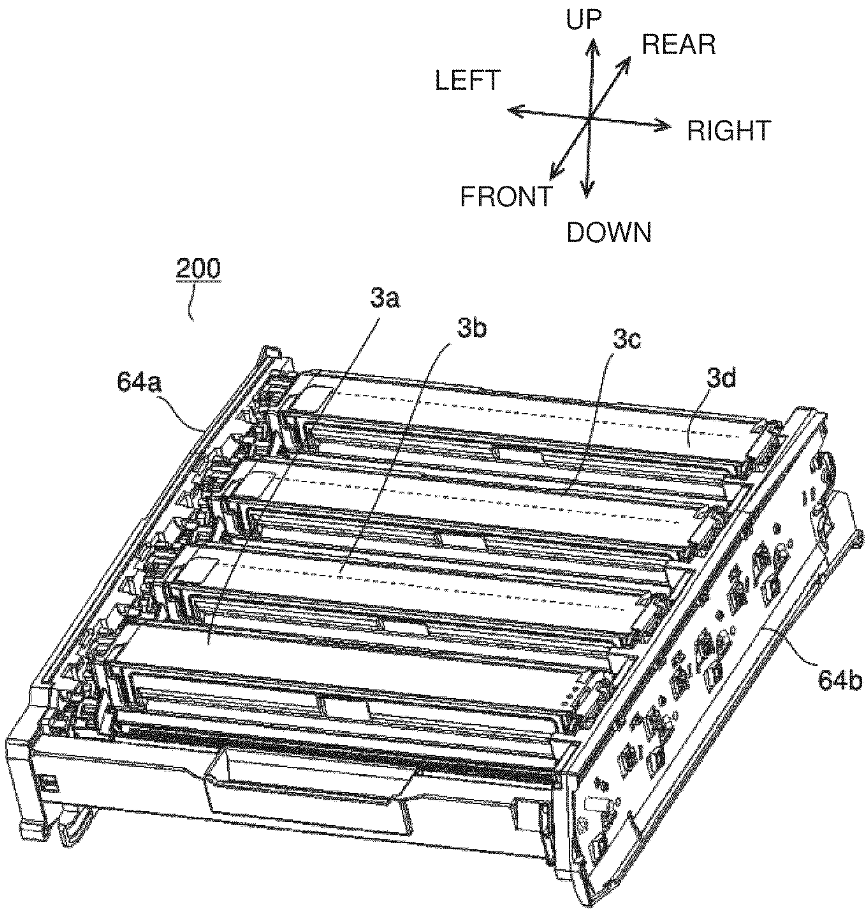


Fig.6

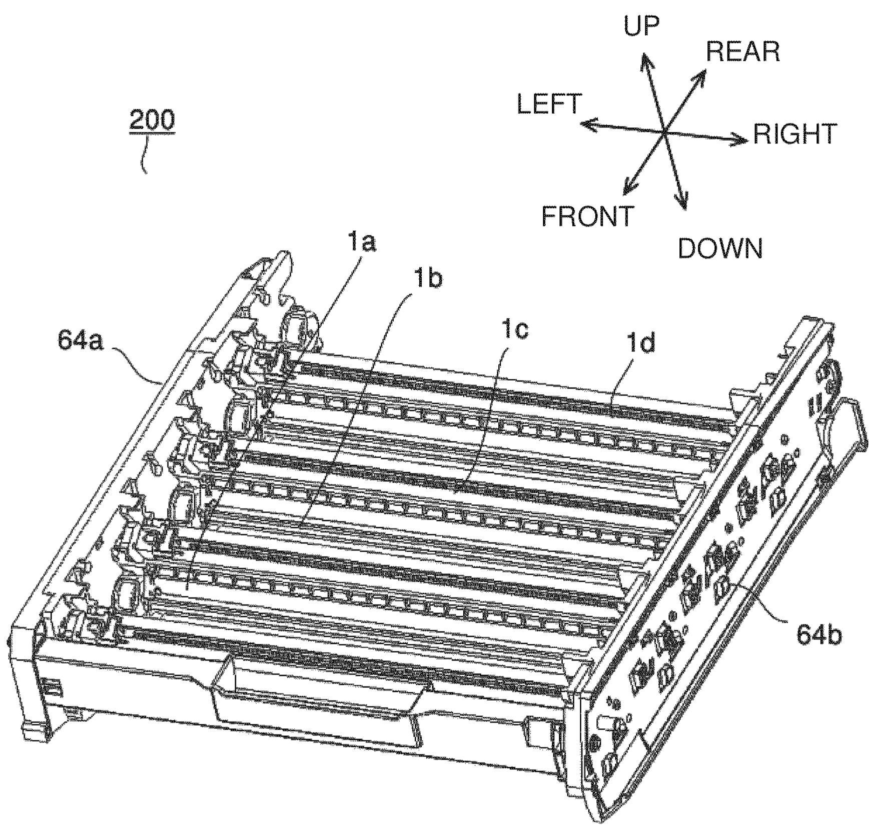
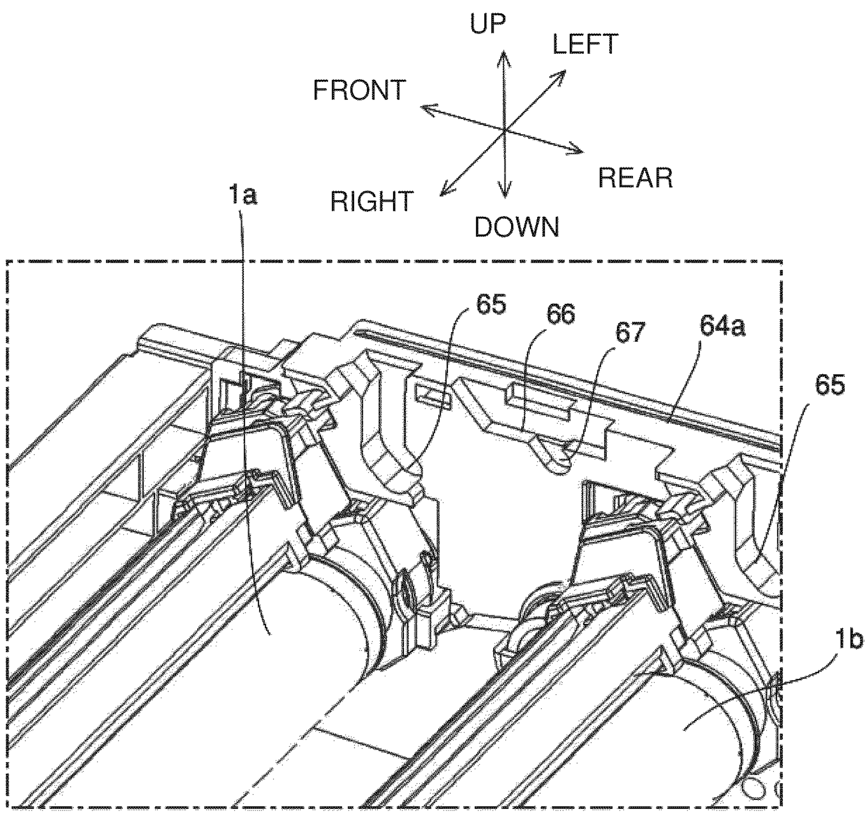


Fig.7



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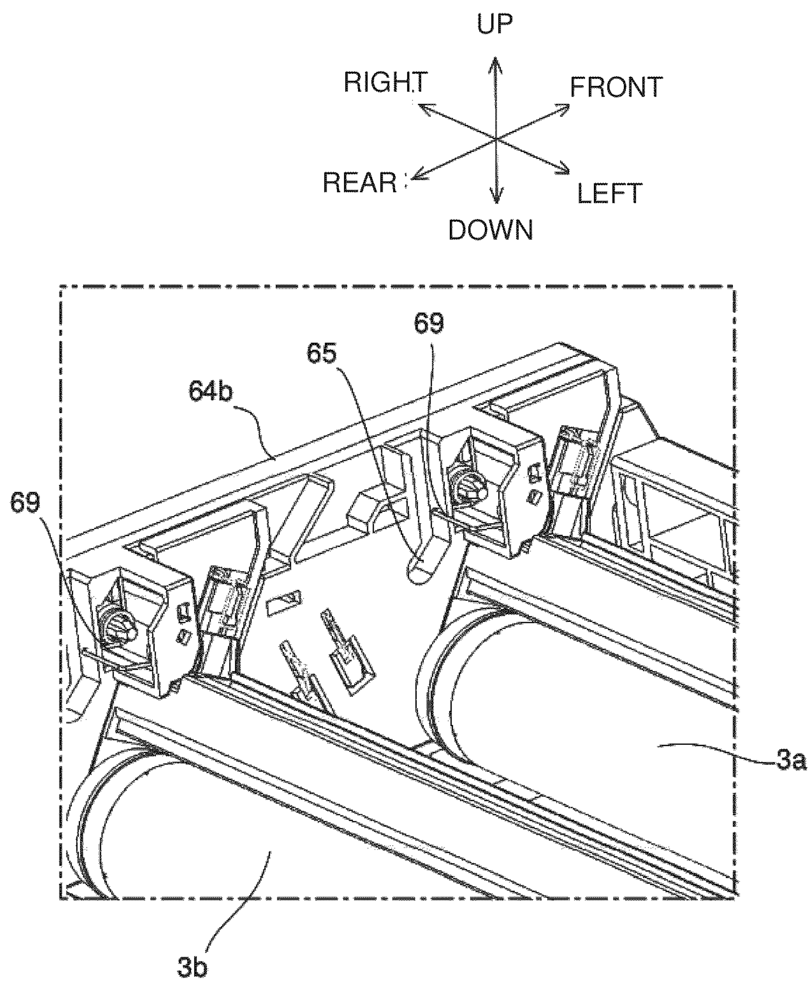


Fig.9

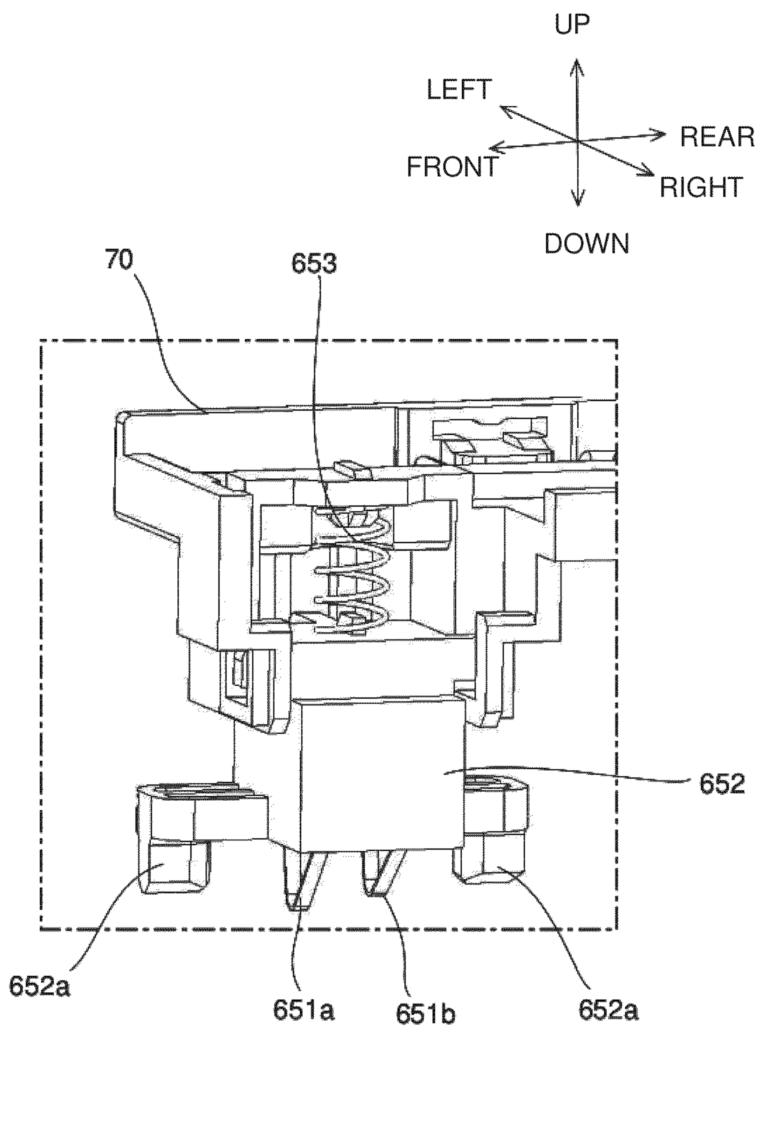
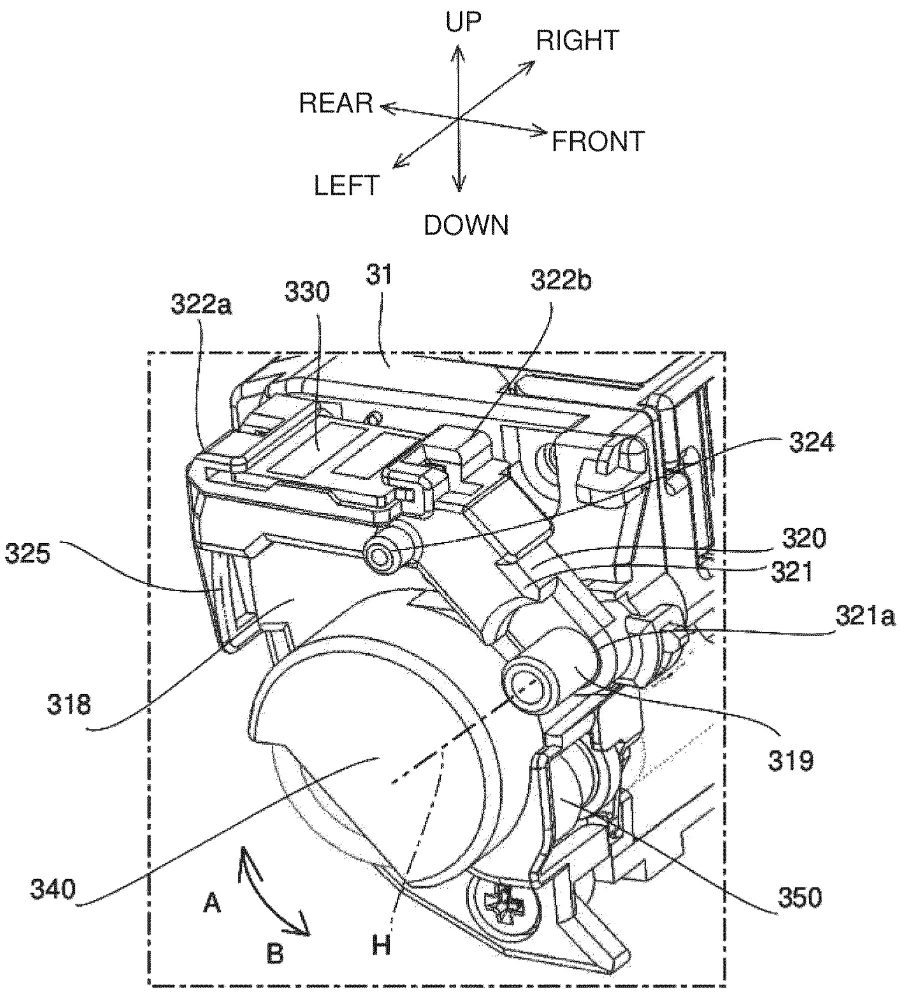


Fig.10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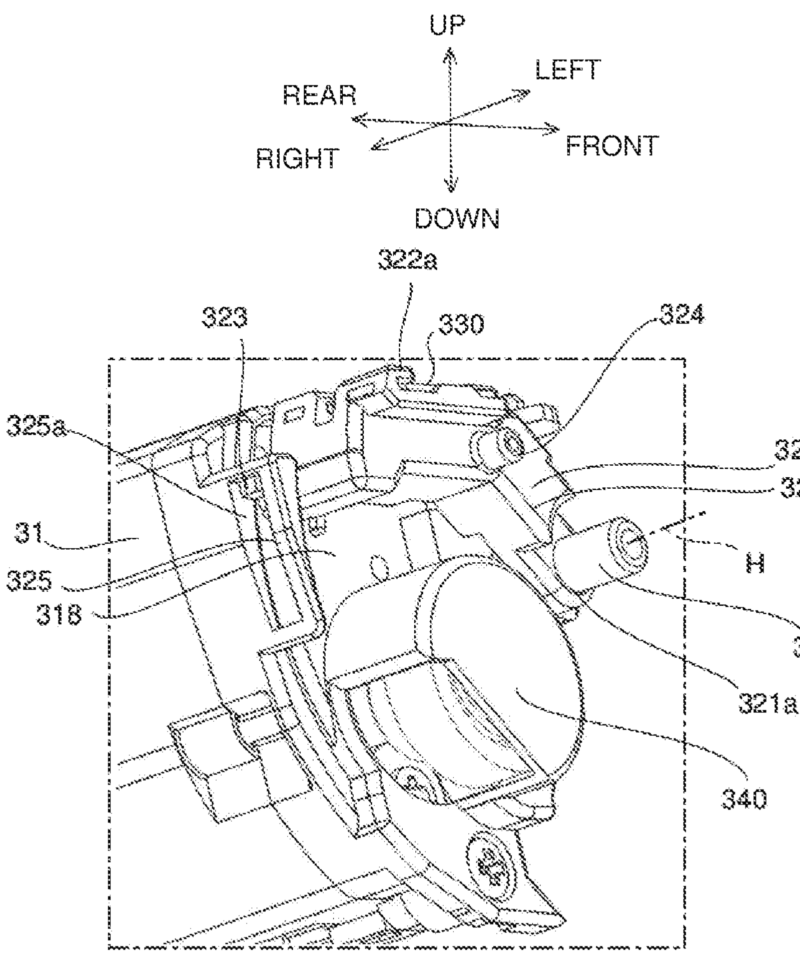
A
B
H

Fig.11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321a
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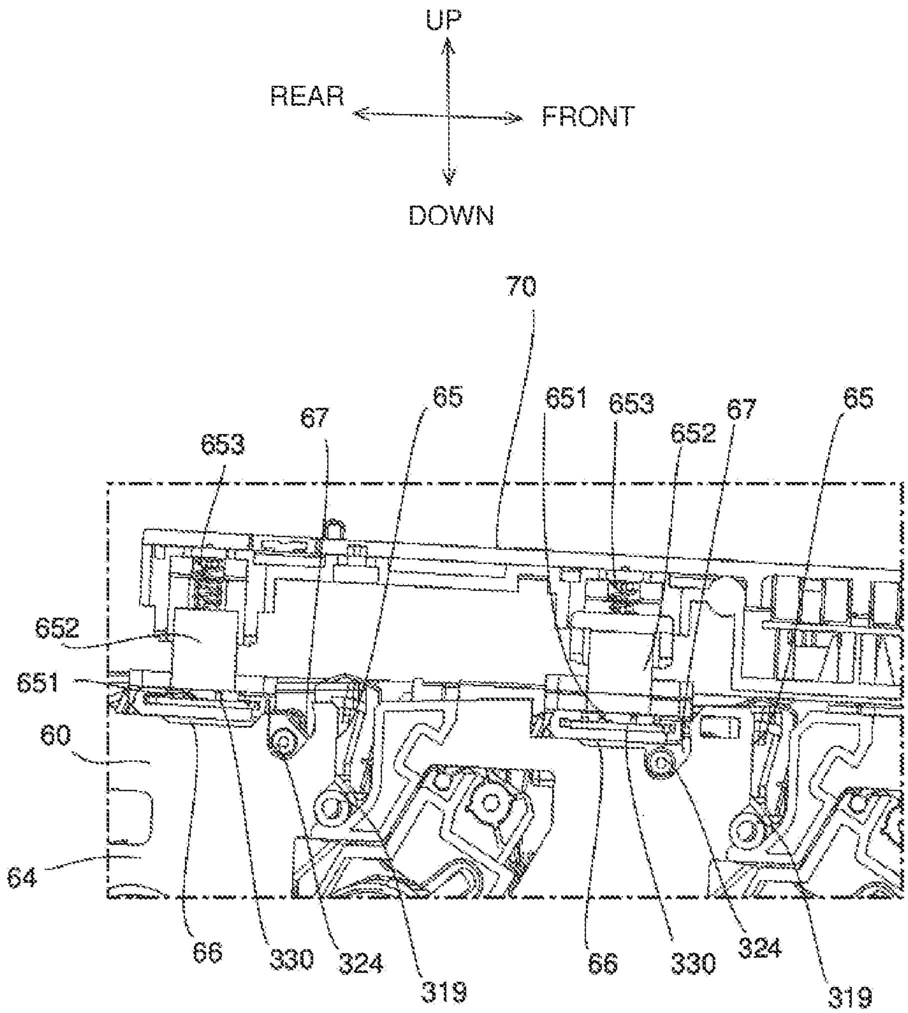
Fig.12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324
319
66
330
324
319

IMAGE FORMING APPARATUS

INCORPORATION BY REFERENCE

This application is based on and claims the benefit of priority from Japanese Patent Application No. 2023-127510 filed on Aug. 4, 2023, the contents of which are hereby incorporated by reference.

BACKGROUND

The present disclosure relates to an image forming apparatus.

A conventional image forming apparatus includes an image carrier, a developing unit, and a housing. The image carrier is subjected to formation of an electrostatic latent image on its surface. The developing unit feeds developer onto the surface of the image carrier, and develops the electrostatic latent image to form a toner image. The housing houses the image carrier and the developing unit therein.

The developing unit includes a toner carrier, a development container, and a circuit board. The toner carrier, placed opposite to the image carrier, carries toner to be fed to the surface of the image carrier. The development container rotatably supports both end portions of a rotating shaft of the toner carrier, and also contains the toner and the toner carrier therein. The circuit board is placed at axial one end portion of the development container.

The developing unit is fitted within the housing, by which the circuit board is connected to a connecting terminal on the housing side.

With the prior art adopted, in a case where the developer consists of nonmagnetic one-component toner, the development container is pivoted on a container support portion serving as a fulcrum so that the toner carrier is reciprocatively moved in a contacting/separating direction relative to the image carrier. In this case, there has been a possibility that the circuit board may be pivoted in conjunction with the development container, resulting in a noncontact state between the circuit board and the connecting terminal.

In view of the above-described problems, an object of the present disclosure is to provide an image forming apparatus capable of maintaining a stable contact between the circuit board and the connecting terminal.

SUMMARY

An image forming apparatus according to one aspect of the present disclosure includes an image carrier, a developing unit, and a housing. The image carrier has an electrostatic latent image formed on its surface. The developing unit feeds nonmagnetic one-component developer to the surface of the image carrier and develops the electrostatic latent image to form a toner image. The housing contains the image carrier and the developing unit. The developing unit includes a toner carrier, a development container, a circuit board, and a board retaining portion. The toner carrier is placed opposite the image carrier, and carries the developer to be fed to the surface of the image carrier. The development container rotatably supports both end portions of a rotating shaft of the toner carrier, and contains the developer and the toner carrier. The circuit board is placed in adjacency to axial one side of the development container. The board retaining portion retains the circuit board. The housing includes one pair of retaining frames, and a terminal holder. The retaining frames retain the development container and the image carrier from both sides of an axial direction. The terminal holder is placed above the retaining frames, and is vertically opposed to the circuit board at one end portion of the axial direction of the development container. The terminal holder includes a connecting terminal, and an elastic member. The connecting terminal is set in contact with the circuit board. The elastic member presses the connecting terminal toward the circuit board. The development container includes a protrusive support portion which protrudes from axial both end portions and which is pivotably supported by the retaining frames. The development container is pivoted on the protrusive support portion serving as a fulcrum, and the toner carrier is reciprocatively movable in such a direction as to come into contact with or separate farther from the image carrier. The board retaining portion is pivotably supported on a fulcrum given by a retention support portion coaxial with the protrusive support portion, and is independent of the reciprocative movement of the development container.

Still further objects of the disclosure as well as concrete advantages obtained by the disclosure will become more apparent from embodiments thereof described hereinbelow.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of an image forming apparatus **100** according to an embodiment of the present disclosure;

FIG. **2** is a side sectional view showing an internal configuration of the image forming apparatus **100** according to the embodiment of the disclosure;

FIG. **3** is a perspective view showing part of the image forming apparatus **100**;

FIG. **4** is a side sectional view showing part of the image forming apparatus **100**;

FIG. **5** is a perspective view showing a drum unit **200** to be mounted on the image forming apparatus **100**;

FIG. **6** is a perspective view showing the drum unit **200** to be mounted on the image forming apparatus **100**;

FIG. **7** is a perspective view showing, in enlargement, part of a retaining frame **64***a* to be mounted on the image forming apparatus **100**;

FIG. **8** is a perspective view showing, in enlargement, part of a retaining frame **64***b* to be mounted on the image forming apparatus **100**;

FIG. **9** is a perspective view showing, in enlargement, part of a terminal holder **70** to be mounted on the image forming apparatus **100**;

FIG. **10** is a perspective view showing, in enlargement, part of an end portion of a developing unit **3***a* to be mounted on the image forming apparatus **100**;

FIG. **11** is a perspective view showing, in enlargement, an end portion of the developing unit **3***a* to be mounted on the image forming apparatus **100**; and

FIG. **12** is a side sectional view showing part of the developing unit **3***a* mounted on the image forming apparatus **100**.

DETAILED DESCRIPTION

(1. General Configuration of Image Forming Apparatus)

Hereinafter, an embodiment of the present disclosure will be described with reference to the accompanying drawings. FIG. **1** is a perspective view of an image forming apparatus **100** according to an embodiment of the disclosure. FIG. **2** is a side sectional view showing an internal structure of the image forming apparatus **100**. It is noted that positioning of the internal structure of the image forming apparatus **100** in FIG. **2** is not limitative in terms of positional relationships of the disclosure.

The image forming apparatus **100** includes photosensitive drums (image carriers) la to **1***d*, developing units **3***a* to **3***d*, and a housing **300**. The housing **300** constitutes a shell of the image forming apparatus **100**, internally accommodating the photosensitive drums (image carriers) la to **1***d*, and the developing units **3***a* to **3***d*.

In a main body of the image forming apparatus **100** (color printer in this case), four image forming parts Pa, Pb, Pc, Pd are disposed in this order starting from an upstream side (left side in FIG. **1**) of a conveyance direction. These image forming parts Pa to Pd, which are provided in correspondence to images of different four colors (yellow, cyan, magenta and black), form images of yellow, cyan, magenta and black successively each through processes of charging, exposure to light, development and transfer.

The image forming parts Pa to Pd include photosensitive drums (image carriers) la to **1***d*, charging units **2***a* to **2***d*, an exposure unit **5**, developing units **3***a* to **3***d*, and cleaning units **7***a* to **7***d*, respectively. The photosensitive drums (image carriers) **1***a* to **1***d* for carrying visible images (toner images) of individual colors, respectively, are disposed in the image forming parts Pa to Pd, respectively, while an intermediate transfer belt (transfer-destination member) **8** is provided in adjacency to the image forming parts Pa to Pd.

Primary transfer rollers (transfer members) **6***a* to **6***d* are placed in opposition to the photosensitive drums (image carriers) **1***a* to **1***d*, respectively. By the primary transfer rollers **6***a* to **6***d*, individual-color visible images (toner images) formed on the photosensitive drums (image carriers) **1***a* to **1***d* with a specified transfer voltage applied thereto are transferred onto the intermediate transfer belt (transfer-destination member) **8**. As a result of this, the toner images formed on the photosensitive drums **1***a* to **1***d* are primarily transferred, and superimposed one on another, successively onto the intermediate transfer belt **8** that is moving while keeping in contact with the photosensitive drums **1***a* to **1***d*.

Thereafter, the toner image primarily transferred onto the intermediate transfer belt **8** is secondarily transferred onto a transfer sheet P as an example of a recording medium by a secondary transfer roller **9**. Further, the transfer sheet P to which the toner image has been secondarily transferred is subjected to fixation of the toner image in a fixing unit **13** and thereafter discharged from the main body of the image forming apparatus **100**. In execution of the image forming process with the photosensitive drums **1***a* to **1***d*, the photosensitive drums **1***a* to **1***d* are each rotated counterclockwise in FIG. **2**.

The transfer sheet P, to which the toner image is to be secondarily transferred, is contained in a sheet cassette **16** placed in lower part of the main body of the image forming apparatus **100**. The transfer sheet P is conveyed via a feed roller **12***a* and a registration roller pair **12***b* to a nip portion between the secondary transfer roller **9** and a driving roller **11** of the intermediate transfer belt **8**. The intermediate transfer belt **8** is mostly given by a seamless belt with use of a sheet of dielectric resin. Also, a blade-like belt cleaner **19** for eliminating toner or the like remaining on the surface of the intermediate transfer belt **8** is placed downstream of the secondary transfer roller **9**.

Next, the image forming parts Pa to Pd will be described. Provided around and above the photosensitive drums (image carriers) **1***a* to **1***d* that are set up rotatable are charging units **2***a* to **2***d* for electrically charging the photosensitive drums **1***a* to **1***d*, respectively, an exposure unit **5** for exposing the photosensitive drums **1***a* to **1***d* to light of image information, developing units **3***a* to **3***d* for forming toner images on the photosensitive drums **1***a* to **1***d*, respectively, and cleaning units **7***a* to **7***d* for eliminating developer (toner) or the like remaining on the photosensitive drums **1***a* to **1***d*, respectively.

The exposure unit **5** exposes to light the photosensitive drums (image carriers) **1***a* to **1***d* that have been electrically charged by the charging units **2***a* to **2***d* to form electrostatic latent images, respectively.

The developing units **3***a* to **3***d* are placed in opposition to the photosensitive drums (image carriers) **1***a* to **1***d*, respectively, and include developing rollers (toner carriers) **21***a* to **21***d*, feed rollers **22***a* to **22***d*, and development containers **4***a* to **4***d*, respectively. The developing units **3***a* to **3***d* apply a specified development voltage to the developing rollers **21***a* to **21***d*, respectively, so that nonmagnetic one-component toner (developer) is deposited on the electrostatic latent images formed on the photosensitive drums **1***a* to **1***d*, respectively, to form a toner image. The developing rollers **21***a* to **21***d* are placed in opposition to the photosensitive drums (image carriers) **1***a* to **1***d*, respectively.

The development containers **4***a* to **4***d* have individual-color nonmagnetic one-component toner of yellow, cyan, magenta and black, respectively, contained therein. Placed inside the development containers **4***a* to **4***d* are developing rollers (toner carriers) **21***a* to **21***d*, feed rollers **22***a* to **22***d*, and stirring paddles (not shown), respectively. The stirring paddles stir the toner in the development containers **4***a* to **4***d*, respectively.

The feed rollers **22***a* to **22***d* are placed in opposition to the developing rollers **21***a* to **21***d*, respectively. The feed rollers **22***a* to **22***d* hold, on their outer circumferential surfaces, the toner contained in the development containers **4***a* to **4***d*, respectively. Also, the feed rollers **22***a* to **22***d* feed the developer held on the outer circumferential surfaces to the developing rollers **21***a* to **21***d*, respectively. In order to move the toner from the feed rollers **22***a* to **22***d* to the developing rollers **21***a* to **21***d*, respectively, a specified feed voltage (DC voltage) is applied to the feed rollers **22***a* to **22***d*.

Upon input of image data from a higher-order device such as a personal computer, first, the surfaces of the photosensitive drums **1***a* to **1***d* are electrically charged uniformly by the charging units **2***a* to **2***d*, respectively. Next, light emission corresponding to the image data is executed by the exposure unit **5**, so that electrostatic latent images corresponding to the image data are formed on the photosensitive drums **1***a* to **1***d*, respectively. The developing units **3***a* to **3***d* have been filled with specified quantities of developer including individual-color toner of yellow, cyan, magenta and black, respectively. The toner in the developer is fed by the developing units **3***a* to **3***d* onto the photosensitive drums **1***a* to **1***d*, respectively, so as to be electrostatically deposited thereon. As a result of this, toner images corresponding to the electrostatic latent images formed by the exposure from the exposure unit **5** are formed. In this embodiment, the developer is a nonmagnetic one-component developer.

Then, by the primary transfer rollers **6***a* to **6***d*, electric fields are imparted at a specified transfer voltage to between the primary transfer rollers **6***a* to **6***d* and the photosensitive drums **1***a* to **1***d*, respectively, so that the toner images of yellow, cyan, magenta and black on the photosensitive drums **1***a* to **1***d* are primarily transferred onto the intermediate transfer belt **8**. These four-color images are formed with specified positional relationships which are previously determined for specified full-color image formation. Thereafter, in preparation for subsequent formation of new electrostatic latent images, toner or the like remaining on the surfaces of the photosensitive drums **1***a* to **1***d* after the primary transfer is eliminated by the cleaning units **7***a* to **7***d*.

The intermediate transfer belt **8** is stretched over on an upstream-side driven roller **10** and a downstream-side driving roller **11**. When the intermediate transfer belt **8** has started to turn clockwise along with rotation of the driving roller **11** driven by a drive motor (not shown), the transfer sheet P is conveyed at a specified timing from the registration roller pair **12***b* to a nip portion (secondary transfer nip portion) between the driving roller **11** and the secondary transfer roller **9** provided in adjacency to the driving roller **11**, so that the full-color image on the intermediate transfer belt **8** is secondarily transferred onto the transfer sheet P. The transfer sheet P to which the toner image has been secondarily transferred is conveyed to the fixing unit **13**.

The transfer sheet P conveyed to the fixing unit **13** is heated and pressurized by a fixing roller pair **13***a* so that the toner image is fixed on the surface of the transfer sheet P, with a specified full-color image formed. The transfer sheet P with the full-color image formed thereon is assorted in conveyance direction by a branching part **14** that branches into a plurality of directions, and the transfer sheet P is discharged, as it is (or after fed to a double-sided conveyance path **18** and subjected to double-sided image formation), to a discharge tray **17** by a discharge roller pair **15**.

(2. Configuration of Drum Unit)

FIG. **3** is a perspective view showing part of the image forming apparatus **100**. FIG. **4** is a side sectional view showing part of the image forming apparatus **100**. FIGS. **3** and **4** show a state in which the drum unit **200** is exposed with the housing **300** partly removed from the image forming apparatus **100**. FIGS. **5** and **6** are perspective views of the drum unit **200**. FIG. **5** shows a state in which the developing units **3***a* to **3***d* have been set up, and FIG. **6** shows a state in which the developing units **3***a* to **3***d* have been removed.

In the image forming apparatus **100**, pulling out a front cover **301**, which is placed at a front of the housing **300**, causes the drum unit **200** to be exposed. In this state, a user is allowed to take out the developing units **3***a* to **3***d* supported by the drum unit **200**.

The drum unit **200** includes the photosensitive drums (image carriers) **1***a* to **1***d*, the developing units **3***a* to **3***d*, the charging units **2***a* to **2***d*, the exposure unit **5**, retaining frames **64***a*, **64***b*, the cleaning units **7***a* to **7***d*, and a terminal holder **70**.

When the developing units **3***a* to **3***d* have become empty of contained developer, the developing units **3***a* to **3***d* can be replaced with other developing units **3***a* to **3***d* filled with the developer. Also, the developing units **3***a* to **3***d* are removed, as needed, for maintenance of interior of the image forming apparatus **100**. In this embodiment, the developing units **3***a* to **3***d* are fittably-and-removably supported on the housing **300**.

The retaining frames **64***a*, **64***b* are part of the housing **300**, and the drum unit **200** is fixed to the housing **300** via the retaining frames **64***a*, **64***b*. That is, the housing **300** includes the retaining frames **64***a*, **64***b*.

The retaining frames **64***a*, **64***b* are metallic plate-shaped members extending in a front/rear direction, and provided in one pair with the photosensitive drums (image carriers) **1***a* to **1***d* and the developing units **3***a* to **3***d* both interposed in a left/right direction between the retaining frames **64***a*, **64***b*. In this case, the photosensitive drums **1***a* to **1***d* and the developing units **3***a* to **3***d* are arrayed alternately in the front/rear direction.

The terminal holder **70** is movable vertically, i.e. in an up/down direction, via a driving unit (not shown). When the front cover **301** is pulled out forward, the terminal holder **70** is moved upward so as to become farther from the drum unit **200**. Conversely, when the front cover **301** is pushed in rearward so as to be fitted to the housing **300**, the terminal holder **70** is moved downward so as to become closer to the drum unit **200**.

FIG. **7** is a perspective view showing, in enlargement, part of the retaining frame **64***a*. FIG. **8** is a perspective view showing, in enlargement, part of the retaining frame **64***b*. The pair of retaining frames **64***a*, **64***b* each have guide portions **65**, board recesses **66**, and restrictive recesses **67**.

The guide portions **65** are placed on inner surfaces of the retaining frames **64***a*, **64***b*, respectively, opposed to each other in the left/right direction. The guide portions **65** are placed at four places in each of the retaining frames **64***a*, **64***b* so as to be arrayed with specified intervals in the front/rear direction. During course of fitting the developing units **3***a* to **3***d* to the drum unit **200**, the guide portions **65** guide the developing units **3***a* to **3***d* to their specified fitting positions. In this embodiment, the guide portions **65** are recesses formed in the inner surfaces of the retaining frames **64***a*, **64***b*, respectively, with their upper end faces opened. Also, each guide portion **65** has its upper end portion extending downward linearly and its lower end portion bent in a direction of further separating from the photosensitive drum **1***a*. In this embodiment, the lower end portion of each guide portion **65** has rearward inclination that increases more and more in a downward direction.

Each board recess **66** is formed so as to be recessed downward from an upper surface of the retaining frame **64***a*, with a later-described circuit board **330** partly placed inside the board recess **66**. The board recess **66** is placed at four places arrayed in the front/rear direction, where the board recesses **66** are, in this embodiment, placed in rearward adjacency to the guide portions **65**, respectively.

Each restrictive recess **67** is formed by a bottom surface of the board recess **66** being partly recessed downward. The restrictive recess **67**, in which a later-described restrictive protrusion **324** is to be placed, differs in shape in response to the restrictive protrusion **324** placed therein. As a result, any one of the developing units **3***a* to **3***d*, whichever it is incompatible with a restrictive recess **67**, is prevented from being fitted, thus the restrictive recess **67** serving for color-incompatibility function.

FIG. **9** is a perspective view showing, in enlargement, part of the terminal holder **70**. The terminal holder **70**, forming part of the housing **300**, is fixed to the housing **300** so as to be movable vertically, i.e., in the up/down direction. That is, the housing **300** includes the terminal holder **70**. More specifically, the terminal holder **70**, extending in the front/rear direction, is movable in the up/down direction above the retaining frame **64***a* placed on the left side (axial one side).

The terminal holder **70** includes a plurality of connecting terminals **651***a*, **651***b*, a terminal retaining portion **652**, and a coil spring (elastic member) **653**. As the connecting terminals **651***a*, **651***b*, the terminal retaining portion **652** and the coil spring (elastic member) **653** are assumed as one set, four sets of these members are placed so as to be arrayed in the front/rear direction in correspondence to the developing units **3***a* to **3***d*, respectively.

The terminal retaining portion **652** retains the connecting terminals **651***a*, **651***b* at lower end portions, respectively, and also has a pair of board pressing pieces **652***a* protruding downward from a lower face of the terminal retaining portion **652**. The connecting terminals **651***a*, **651***b* are placed between a pair of board pressing pieces **652***a*. The pair of board pressing pieces **652***a* are so positioned that their lower ends become higher than lower ends of the connecting terminals **651***a*, **651***b*.

When the terminal holder **70** is moved downward so as to be close to the drum unit **200**, the connecting terminals **651***a*, **651***b* are brought into contact, and connected, with an upper face of a later-described circuit board **330**. The connecting terminals **651***a*, **651***b* are biased downward via the coil spring **653** that presses downward an upper face of the terminal retaining portion **652**. That is, the coil spring (elastic member) **653** presses the connecting terminals **651***a*, **651***b* toward the circuit board **330**.

The connecting terminals **651***a*, **651***b* biased downward are brought into contact, and deformed, with the circuit board **330**. As a result, the circuit board **330** and the connecting terminals **651***a*, **651***b* come into a stably contactable state. In this case, the board pressing pieces **652***a* are brought into contact with the circuit board **330**, so that the upper face of the circuit board **330** and the lower face of the terminal retaining portion **652** are kept constant in vertical distance therebetween. Thus, the circuit board **330** and the connecting terminals **651***a*, **651***b* come into a more stably contactable state therebetween.

The connecting terminals **651***a*, **651***b* are elastically deformable and moreover different therebetween in lower end position in the vertical direction, i.e. up/down direction. In this embodiment, the lower end of the connecting terminal **651***a* is positioned lower than the lower end of the connecting terminal **651***b*. Also, the connecting terminal **651***a* is assumed as a grounding-dedicated terminal, and the connecting terminal **651***b* is assumed as an electrical conduction- and signal-dedicated terminal. With this assumption, when the terminal holder **70** has moved downward so as to be close to the drum unit **200**, the connecting terminal **651***a* comes into contact with the circuit board **330** earlier than the connecting terminal **651***b*. Therefore, in replacement of the circuit board **330**, the circuit board **330** can be protected from a hot-swapping state and thereby prevented from being damaged.

By the connecting terminals **651***a*, **651***b* being connected to the circuit board **330**, the developing unit **3***a* is electrically connected to a main control unit (not shown) consisting of a CPU and the like for controlling individual units of the image forming apparatus **100**.

(3. Configuration of Set-and-Removal Mechanism for Developing Unit)

FIGS. **10** and **11** are perspective views showing, in enlargement, a left-side (one-side) end portion of the developing unit **3***a*. Axial both side faces of the development container **31** are covered with side wall portions **318**, respectively, where each side wall portion **318** includes bearing portions **340**, **350** and a protrusive support portion **319**. The bearing portion **340** rotatably supports a developing roller **32**. The bearing portion **350** rotatably supports a feed roller **33**. With this configuration, the developing roller **32** and the feed roller **33** are supported rotatable in the development container **31**.

The protrusive support portions **319** protrude from side faces of both side wall portions **318** axially outward so as to be inserted into the above-described guide portions **65**, respectively. The development container **31** is supported rotatable by the protrusive support portions **319**.

Further, the circuit board **330** and a board retaining portion **320** for retaining the circuit board **330** are placed axially outside of the left-side (axial one-side) side wall portion **318** of the development container **31**. That is, the circuit board **330** is placed in left-side (axial one-side) adjacency to the development container **31**. The board retaining portion **320** has an arm portion **321**, claw portions **322***a*, **322***b*, a retaining protrusion **323**, and a restrictive protrusion **324**.

The arm portion **321** extends radially from a pivotal shaft H while its fore end portion is bent circumferentially so as to be close to the developing roller **32**. The fore end portion of the arm portion **321** has an upper face formed into a flat shape, where the circuit board **330** is placed.

At a pivotal shaft H-side end portion of the arm portion **321**, a through hole **321***a* is formed so as to run through along the pivotal shaft H. The protrusive support portion **319** is inserted into the through hole **321***a*. As a result of this, the arm portion **321** is supported pivotable on the through hole (retention support portion) **321***a* serving as a fulcrum.

The claw portions **322***a*, **322***b* are bent upward from the upper face of the fore end portion of the arm portion **321** in such a direction as to approach each other. The circuit board **330** placed on the upper face of the fore end portion of the arm portion **321** is pinched and fixed between the claw portions **322***a*, **322***b* and the upper face of the fore end portion of the arm portion **321**.

The retaining protrusion **323** is formed so as to protrude radially outward of the pivotal shaft H from the fore end portion of the arm portion **321**. The retaining protrusion **323** is placed within a slit **325***a* which is formed in a protrusion guide portion **325** protruding axially from an outer circumferential edge of the side wall portion **318**. The slit **325***a* extends circumferentially by running through the protrusion guide portion **325** along the radial direction of the pivotal shaft H. By the retaining protrusion **323** being moved along the slit **325***a*, the board retaining portion **320** is enabled to pivot stably without shifting in the axial direction of the pivotal shaft H.

The restrictive protrusions **324** are inserted into the above-described restrictive recesses **67**, respectively. The restrictive protrusions **324** differ in shape among themselves depending on the developing units **3***a* to **3***d*. As a result of this, the developing units **3***a* to **3***d* can be prevented from being erroneously fitted to the photosensitive drums **1***a* to **1***d* corresponding to wrong colors.

FIG. **12** is a side sectional view showing part of the developing unit **3***a* mounted on the image forming apparatus **100**. The developing units **3***a* to **3***d* are guided along the individual guide portions **65** starting at opening upper ends of those guide portions **65**, respectively. When the developing units **3***a* to **3***d* are guided up to their fitting positions that are terminal ends of the guide portions **65**, respectively, the developing roller **32** is brought into contact with the photosensitive drums **1***a* to **1***d*.

Under this situation, out of the fore end portion of the arm portion **321**, the upper end portion of the arm portion **321** with the circuit board **330** placed on its upper face is mounted on the board recess **66**, and part of the circuit board **330** is placed within the board recess **66**.

When the terminal holder **70** is moved downward so as to be close to the drum unit **200**, the upper face of the circuit board **330** is pressed downward by the connecting terminals **651***a*, **651***b*. As a result of this, the board retaining portion **320** is pinched between the terminal holder **70** and the retaining frame so as to be held at a specified pivotal position. In this case, the circuit board **330** and the connecting terminals **651***a*, **651***b* are brought into contact with each other.

Meanwhile, the development container **31** is pressed by a pressing mechanism **36** (see FIG. **3**) in such a direction as to become closer to the photosensitive drum **1***a*, and the development container **31** is pivoted in such a direction that the developing roller **32** becomes closer to the photosensitive drum **1***a* (A direction in FIG. **10**). In this case, the board retaining portion **320** is supported pivotable on a fulcrum given by the through hole (retention support portion) **321***a* coaxial with the protrusive support portion **319**, independently of reciprocative movement of the development container **31**. As a result of this, the board retaining portion **320** is prevented from pivoting in conjunction with pivoting of the development container **31**, so that the circuit board **330** and the connecting terminals **651***a*, **651***b* are held stably in contact with each other. Accordingly, there can be provided an image forming apparatus **100** capable of keeping a stable contact between the circuit board **330** and the connecting terminals **651***a*, **651***b*.

Also, in a case where the image forming apparatus **100** further includes a pivoting mechanism (not shown) capable of driving the development container **31** into pivotal motion, the development container **31** can be pivoted at an end of printing so as to make the developing roller **32** separate farther from the photosensitive drum **1***a*. In this case, the development container **31** is pivoted in such a direction as to make the developing roller **32** separate farther from the photosensitive drum **1***a* (B direction in FIG. **10**). That is, the development container **31** is pivoted on the protrusive support portion **319** serving as a fulcrum, so that the developing roller **32** can be reciprocatively moved in such a direction as to become closer to or farther from the photosensitive drum **1***a*. In this case, the board retaining portion **320** is supported pivotable on the through hole (retention support portion) serving as a fulcrum and being coaxial with the protrusive support portion **319**, independently of the reciprocative movement of the development container **31**. As a result of this, the board retaining portion **320** can be prevented from being pivoted in conjunction with pivoting of the development container **31**, so that the circuit board **330** and the connecting terminals **651***a*, **651***b* can be held in a stable contact state therebetween.

For instance, during monochromatic printing, the developing roller **32** of the developing unit **3***d* which has been filled with black developer is moved so as to be closer to the photosensitive drum **1***d*, while the developing rollers **32** of the developing units **3***a* to **3***c* which have been filled with developer of yellow, cyan, and magenta are moved so as to be farther from the photosensitive drums **1***a* to **1***c*, respectively. As a result of this, during monochromatic printing, the developing units **3***a* to **3***c* can be kept from being driven, so that drive time of the developing units **3***a* to **3***c* can be reduced. Also, the developing rollers **32** are separated farther from the photosensitive drums **1***a* to **1***c*, respectively, so that cleaning performance of the photosensitive drums **1***a* to **1***d* can be improved.

In addition, this disclosure is not limited only to the above-described embodiment and may be changed and modified in various ways unless those changes and modifications depart from the gist of the disclosure.

The disclosure is applicable to image forming apparatuses using nonmagnetic one-component developer containing toner.

What is claimed is:

1. An image forming apparatus comprising:

an image carrier on a surface of which an electrostatic latent image is to be formed;

a developing unit for feeding nonmagnetic one-component developer to the surface of the image carrier and developing the electrostatic latent image to form a toner image; and

a housing for containing the image carrier and the developing unit, wherein

the developing unit includes:

a toner carrier which is placed opposite the image carrier and which carries the developer to be fed to the surface of the image carrier;

a development container which rotatably supports both end portions of a rotating shaft of the toner carrier and which contains the developer and the toner carrier;

a circuit board which is placed in adjacency to axial one side of the development container; and

a board retaining portion which retains the circuit board, and

the housing includes:

one pair of retaining frames which retain the development container and the image carrier from both sides of an axial direction; and

a terminal holder which is placed above the retaining frames and which is vertically opposed to the circuit board at one end portion of the axial direction of the development container, and

the terminal holder includes:

a connecting terminal which is set in contact with the circuit board; and

an elastic member which presses the connecting terminal toward the circuit board, and

the development container includes

a protrusive support portion which protrudes from axial both end portions and which is pivotably supported by the retaining frames, where

the development container is pivoted on the protrusive support portion serving as a fulcrum, and the toner carrier is reciprocatively movable in such a direction as to come into contact with or separate farther from the image carrier, and

the board retaining portion is pivotably supported on a fulcrum given by a retention support portion coaxial with the protrusive support portion, and is independent of the reciprocative movement of the development container.

2. The image forming apparatus according to claim **1**, wherein

the board retaining portion has, at one end portion, a through hole into which the protrusive support portion is inserted, and also has, on an upper face of the other end portion, an arm portion on which the circuit board is mounted.

3. The image forming apparatus according to claim **2**, wherein

a board recess which is formed by an upper face of the retaining frame being recessed downward, and in which part of the circuit board is placed; and

a restrictive recess which is formed by part of a bottom face of the board recess being recessed downward, where

the restrictive recess accommodates a restrictive protrusion protruding from the board retaining portion, and varies in shape in response to the restrictive protrusion placed therein.

4. The image forming apparatus according to claim **2**, wherein the connecting terminal is elastically deformable and plurally provided and placed, and

the connecting terminals differ in vertical lower-end position thereamong.

5. The image forming apparatus according to claim **1**, wherein

each retaining frame includes:

a board recess which is formed by an upper face of the retaining frame being recessed downward, and in which part of the circuit board is placed; and

a restrictive recess which is formed by part of a bottom face of the board recess being recessed downward, where

the restrictive recess accommodates a restrictive protrusion protruding from the board retaining portion, and varies in shape in response to the restrictive protrusion placed therein.

6. The image forming apparatus according to claim **1**, wherein

the connecting terminal is elastically deformable and plurally provided and placed, and

the connecting terminals differ in vertical lower-end position thereamong.

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